Arunabha Majumder

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Portfolio: arunabha22.github.io/arunabha-majumder

Siemens

Test Design Engineer

Dear Siemens hiring team,

I'd like to apply for the Test Design Engineer role. Designing how a product should be tested, building the setup and then trusting the data that comes out of it is a big part of what I already do, and I would love to do it at Siemens.

As a research assistant at Aalborg University I designed a hybrid shoulder exoskeleton with a variable stiffness mechanism, pretensioned by a worm gear, working in parallel with a 6 Nm motor. I designed it in SolidWorks, 3D printed prototypes, had the final parts manufactured, integrated the motor, encoder and force/torque sensors, and measured its stiffness on a physical test setup. It uses over 25% less motor energy and won the Best Paper Award at ISRM 2026.

I also designed and modelled a novel 5-bar planar parallel robot with variable stiffness mechanisms (VSM), estimated its stiffness with less than 10% error and quantified human arm impedance. For that study with 6 participants, Qualisys cameras recorded the arm position data, and I implemented an Arduino-based trigger so the robot's motion and the recording stayed in sync. In a separate exoskeleton study, I collected EMG data from 12 participants. These projects taught me how much careful test setup, sensor choice and documentation matter.

In my Master's I worked in a team of 4 electronics and mechanical engineers on a soft exosuit, and designed the outer PID loop, with integral anti-windup and error nonlinearisation, of a neural network-based gain-scheduled controller in MATLAB/Simulink and LabVIEW. I also hold a Senior Diploma in Fine Arts and do graphic design (Illustrator, Photoshop, Procreate), which helps me make clear test reports and figures.

I would love to bring new and innovative ideas to your test team and grow with Siemens. My portfolio, with videos and results from my work, is at arunabha22.github.io/arunabha-majumder. Thank you for reading.

Best,
Arunabha Majumder